

07 System of Linear Equations

7.2 Linear Systems Through the Lens of Linear Combination

The Third Perspective of Matrix Multiplication: Linear Combinations, Basis, Collinearity, and Coplanarity



Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Master

1

Revisit the third perspective of matrix multiplication: $\mathbf{Ax}=\mathbf{b}$ as a linear combination of $\mathbf{A}$'s columns

2

Re-analyze the chicken-rabbit puzzle through this lens, understanding “basis” and “coordinates” geometrically

3

See how the identity, diagonal, and triangular matrices shape the grid spanned by the basis

4

Grasp that a zero determinant means a linearly dependent basis, leading to no solution or infinitely many

5

Understand how the solution structure of $\mathbf{Ax}=\mathbf{0}$ relates to linear independence of the columns

6

Meet overdetermined systems: more equations than unknowns, and the three possible solution outcomes

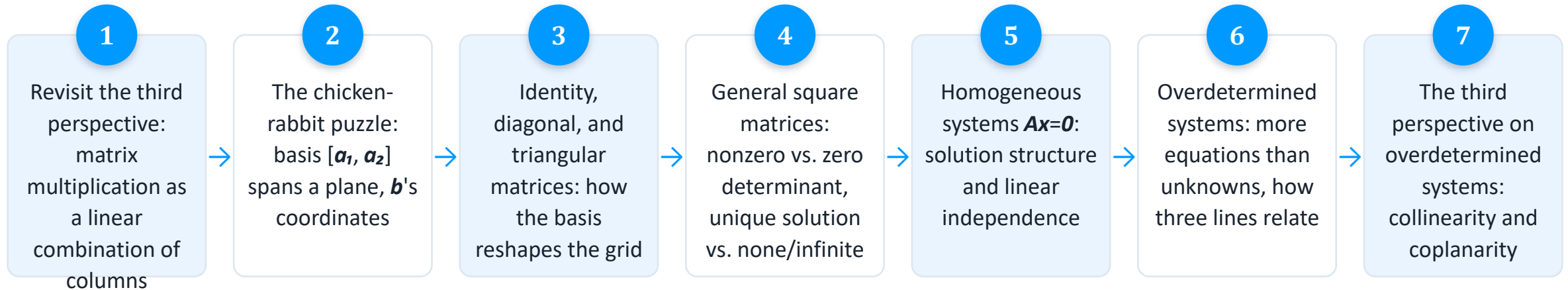
7

Use collinearity and coplanarity to judge whether an overdetermined system has a solution

8

Verify every case in this section with Python — rank, determinant, and least-squares tools from numpy

From Linear Combinations to Overdetermined Systems



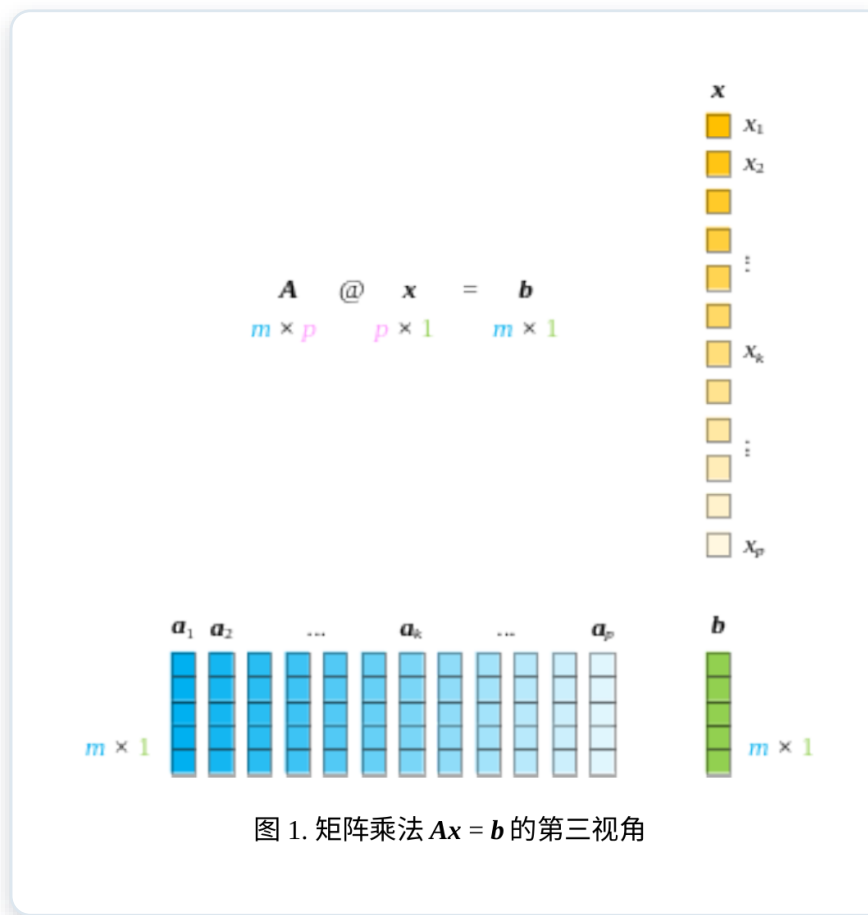
$A\mathbf{x}=\mathbf{b}$ as a Linear Combination of Columns

$$x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \dots + x_p\mathbf{a}_p = \mathbf{b}$$

- Writing $\mathbf{A}$'s columns as $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_p$, $\mathbf{Ax}=\mathbf{b}$ means $\mathbf{b}$ is a linear combination of $\mathbf{A}$'s columns
- The coefficients of that linear combination are exactly the unknown vector $\mathbf{x}=[x_1, x_2, \dots, x_p]^T$

Fig 1 · The Third Perspective of $Ax=b$

- Each column $a_1 \dots a_p$ of A is $m \times 1$; multiplied by x_k and summed term by term, they produce b



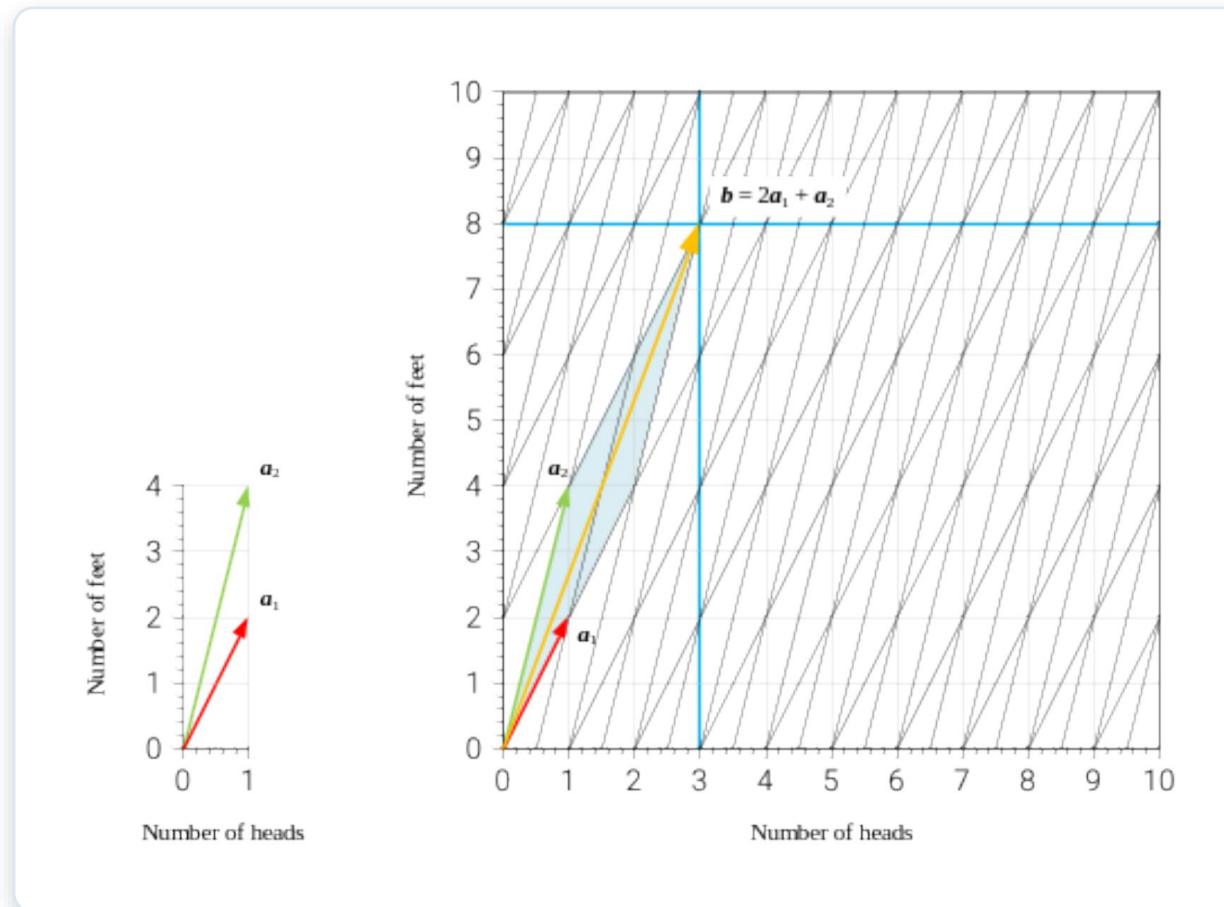
a_1 and a_2 Represent “One Chicken” and “One Rabbit”

$$2a_1 + 1a_2 = b \Rightarrow b = [3, 8]^T$$

- $a_1=[1, 2]^T$: one chicken has 1 head, 2 feet; $a_2=[1, 4]^T$: one rabbit has 1 head, 4 feet
- A simpler example: 2 chickens, 1 rabbit — the head and foot totals are exactly a (2, 1)-combination of a_1, a_2

Fig 2 · $\mathbf{b}$ Is a Linear Combination of $\mathbf{a}_1$ and $\mathbf{a}_2$

- Basis $[\mathbf{a}_1, \mathbf{a}_2]$ spans the whole plane; $\mathbf{b}$ lies in it, with coordinates (2, 1) under this basis



A IS THE IDENTITY MATRIX

$$A=I: a_1=(1, 0), a_2=(0, 1)$$

$$1 \times a_1 - 1 \times a_2 = b \Rightarrow b = [1, -1]^T$$

- Basis $[a_1, a_2]$ spans a unit square grid; b 's coordinates under this basis are $(1, -1)$

Fig 3 · Basis Spans a Unit Square

- a_1 coincides with e_1 , a_2 with e_2 ; the grid is neither scaled nor sheared

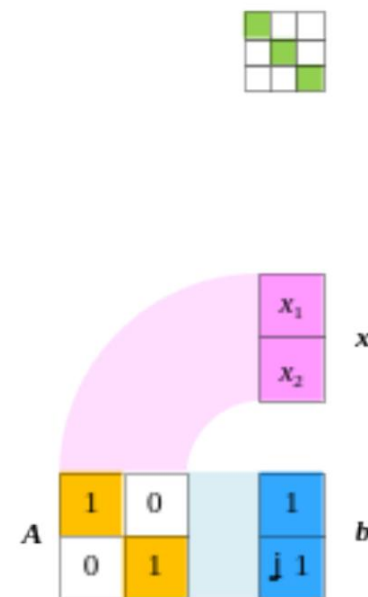
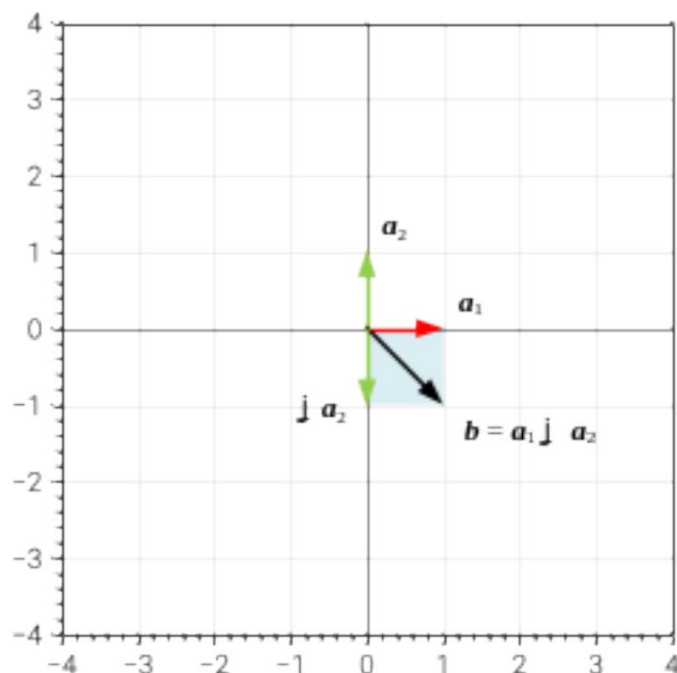


图 3. 方阵 A 是单位矩阵

A IS DIAGONAL · EQUAL ENTRIES

$$\mathbf{a}_1 = (2, 0), \mathbf{a}_2 = (0, 2)$$

$$1 \times \mathbf{a}_1 - 1 \times \mathbf{a}_2 = \mathbf{b} \Rightarrow \mathbf{b} = [2, -2]^T$$

- Basis $[\mathbf{a}_1, \mathbf{a}_2]$ spans a square scaled by 2; $\mathbf{b}$'s coordinates are still $(1, -1)$

Fig 4 · Basis Spans a Uniformly Scaled Square

- The grid grows larger but stays a square — both axes scale by the same factor

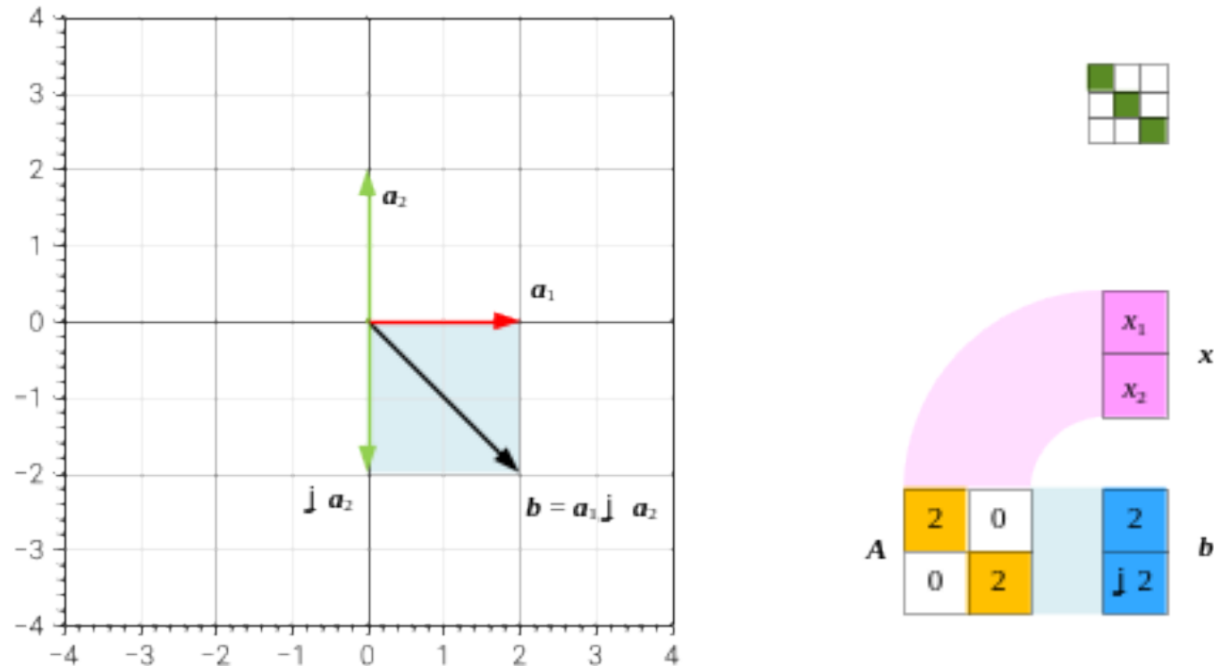


图 4. 方阵 A 是对角方阵，主对角线元素相同

A IS DIAGONAL · UNEQUAL ENTRIES

$$\mathbf{a}_1 = (2, 0), \mathbf{a}_2 = (0, 3)$$

$$1 \times \mathbf{a}_1 - 1 \times \mathbf{a}_2 = \mathbf{b} \Rightarrow \mathbf{b} = [2, -3]^T$$

- Basis $[\mathbf{a}_1, \mathbf{a}_2]$ spans a rectangle; $\mathbf{b}$'s coordinates are still $(1, -1)$

Fig 5 · Basis Spans a Rectangular Grid

- The two axes scale differently (2× and 3×), turning the square grid into a rectangle

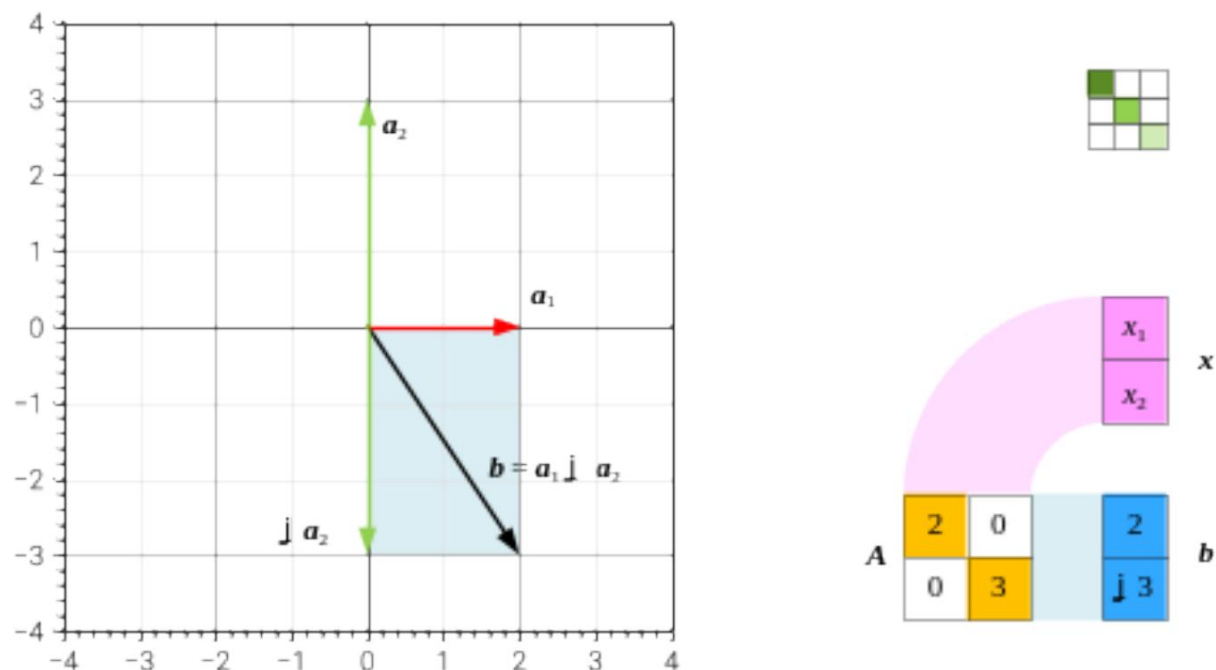


图 5. 方阵 A 是对角方阵，主对角线元素不同

A IS UPPER TRIANGULAR

$$\mathbf{a}_1 = (1, 0), \mathbf{a}_2 = (1, 1)$$

$$1 \times \mathbf{a}_1 - 1 \times \mathbf{a}_2 = \mathbf{b} \Rightarrow \mathbf{b} = [0, -1]^T$$

- Basis $[\mathbf{a}_1, \mathbf{a}_2]$ shears the grid along the horizontal axis; $\mathbf{b}$'s coordinates are still $(1, -1)$

A IS UPPER TRIANGULAR

Fig 6 · Basis Sheared Along the Horizontal Axis

- a_2 tilts to the right, turning the square grid into a parallelogram along the x-axis

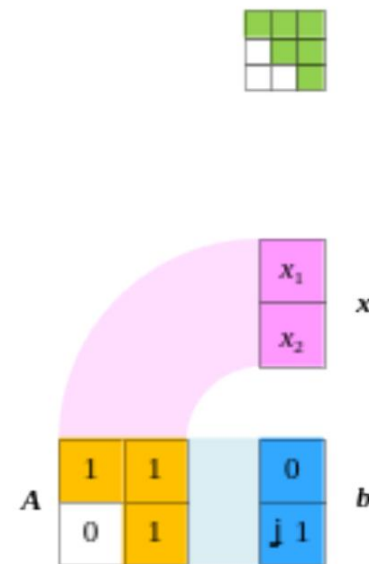
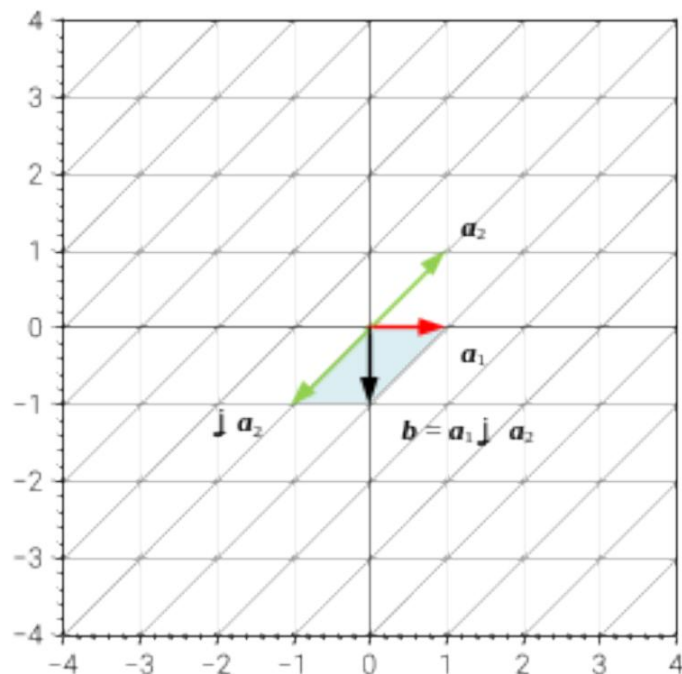


图 6. 方阵 A 是上三角矩阵

A IS LOWER TRIANGULAR

$$\mathbf{a}_1 = (1, 1), \mathbf{a}_2 = (0, 1)$$

$$1 \times \mathbf{a}_1 - 1 \times \mathbf{a}_2 = \mathbf{b} \Rightarrow \mathbf{b} = [1, 0]^T$$

- Basis $[\mathbf{a}_1, \mathbf{a}_2]$ shears the grid along the vertical axis; $\mathbf{b}$'s coordinates are still $(1, -1)$

A IS LOWER TRIANGULAR

Fig 7 · Basis Sheared Along the Vertical Axis

- a_1 tilts upward, turning the square grid into a parallelogram along the y-axis

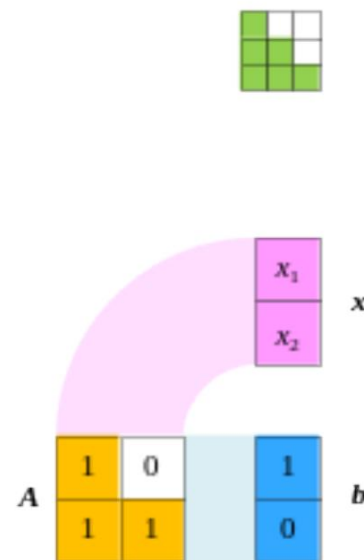
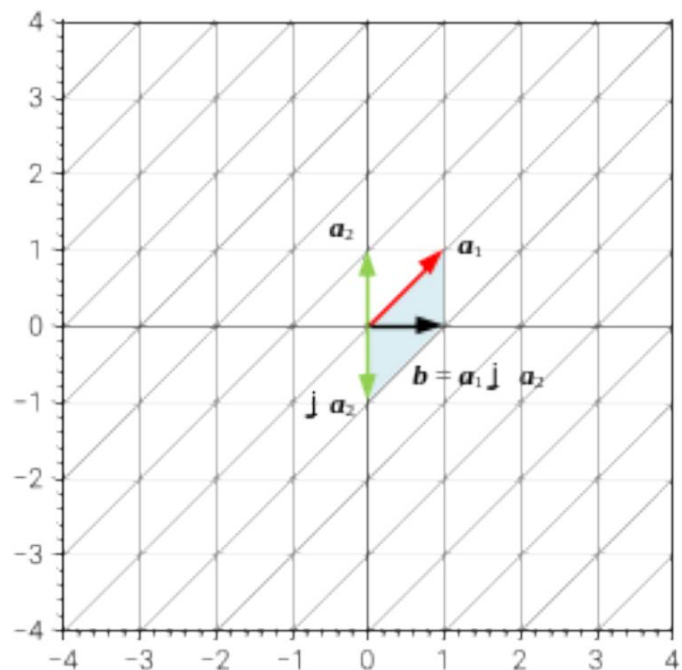


图 7. 方阵 A 是下三角矩阵

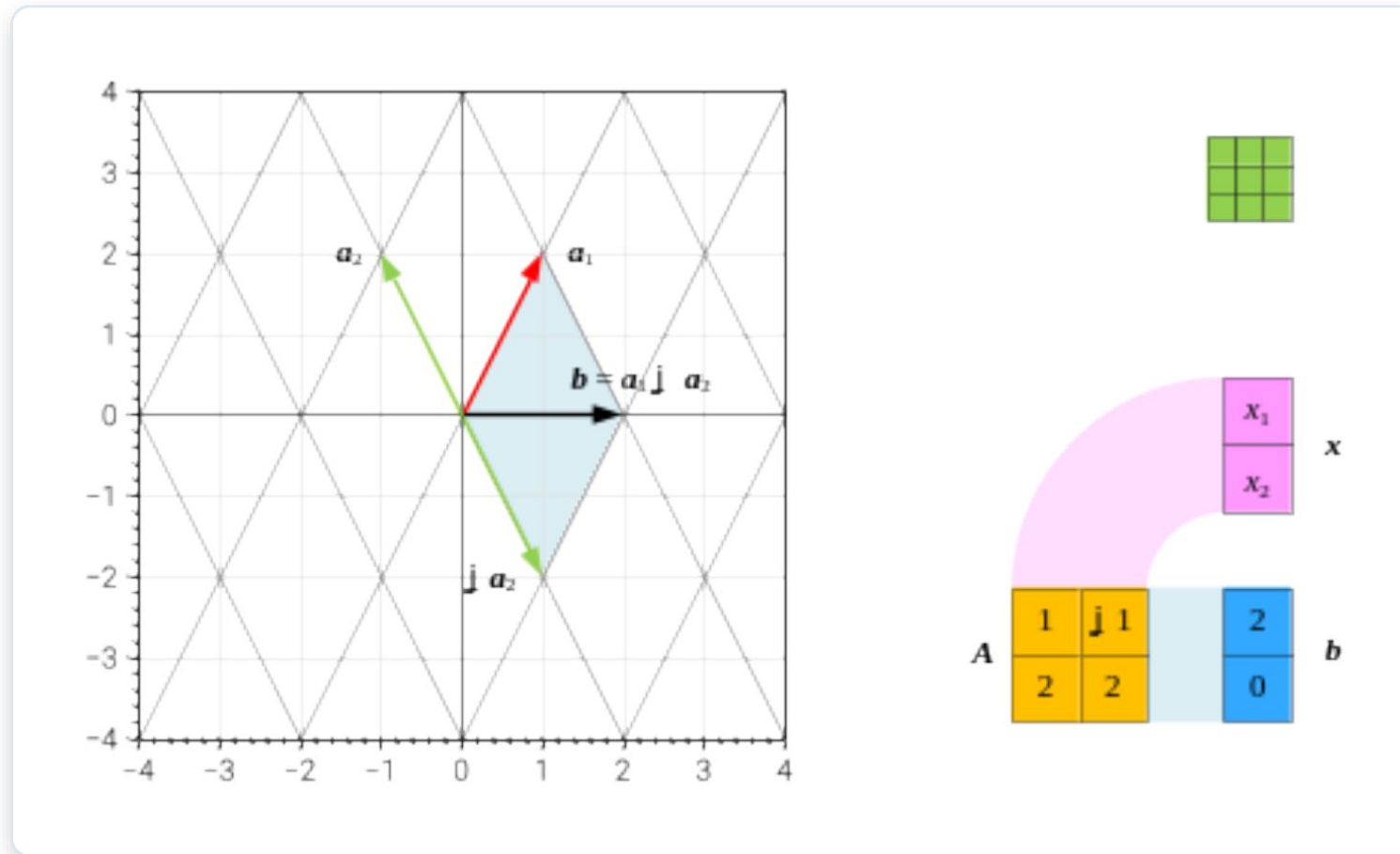
$$\mathbf{a}_1 = (1, 2), \mathbf{a}_2 = (-1, 2)$$

$$1 \times \mathbf{a}_1 - 1 \times \mathbf{a}_2 = \mathbf{b} \Rightarrow \mathbf{b} = [2, 0]^T$$

- $\mathbf{a}_1, \mathbf{a}_2$ are linearly independent, $\det(\mathbf{A})=4 \neq 0$, the basis spans the whole plane; coordinates still $(1, -1)$

Fig 8 · Basis Spans a Diamond Grid, Unique Solution

- a_1, a_2 are neither collinear nor orthogonal, so the grid becomes a general parallelogram



Same A , b Replaced by $\mathbf{0}$

$$A\mathbf{x} = \mathbf{0} \Rightarrow \mathbf{x} = [0, 0]^T$$

- When $\mathbf{a}_1, \mathbf{a}_2$ are linearly independent, the zero vector $\mathbf{0}$'s coordinates can only be $(0, 0)$ — the unique homogeneous solution

Fig 9 · Same Grid as Fig 8, Solution at the Origin

- When $\det(\mathbf{A}) \neq 0$, the homogeneous system $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution $\mathbf{x} = [0, 0]^T$

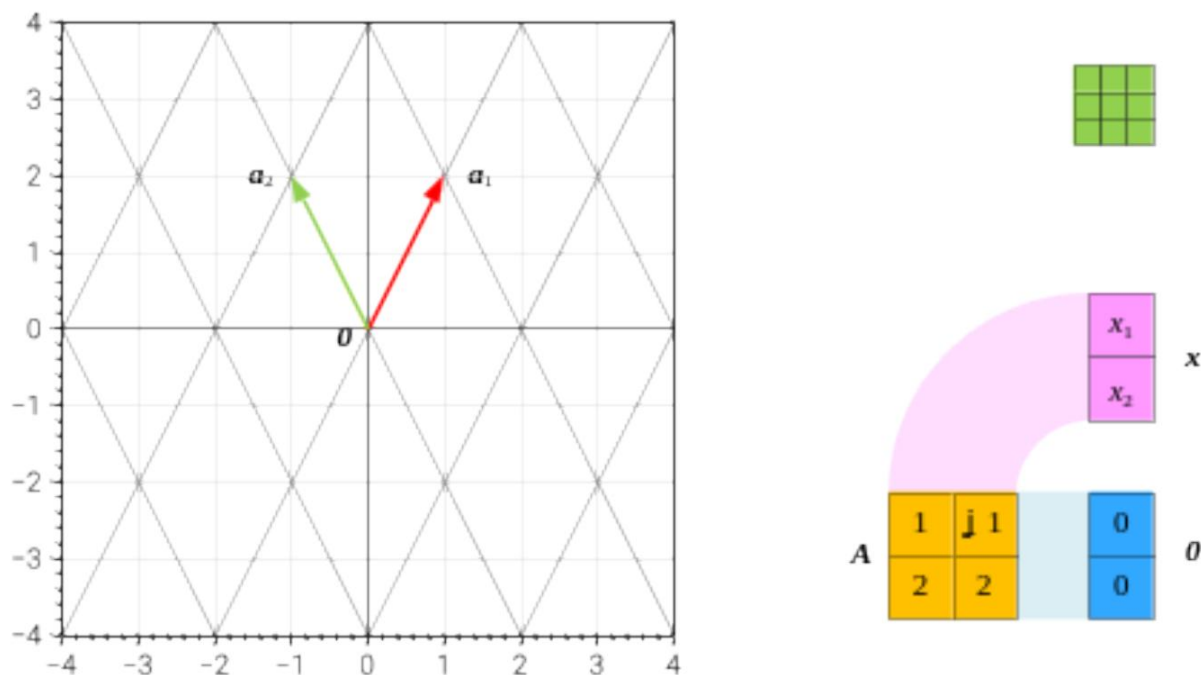


图 9. 方阵 $\mathbf{A}$ 行列式不为 0, $\mathbf{Ax} = \mathbf{0}$

$a_1=(1, 2), a_2=(1, 2)$ Are Collinear

$$k \times a_1 = b \Rightarrow b = [-1, -2]^T$$

- a_1, a_2 are collinear and cannot form a basis; but b also lies on this line, giving infinitely many solutions $x_1=k, x_2=1-k$

Fig 10 · b Lies on the Line Spanned by a_1, a_2

- a_1, a_2 are linearly dependent (collinear), and b also lies on that same line — infinitely many solutions

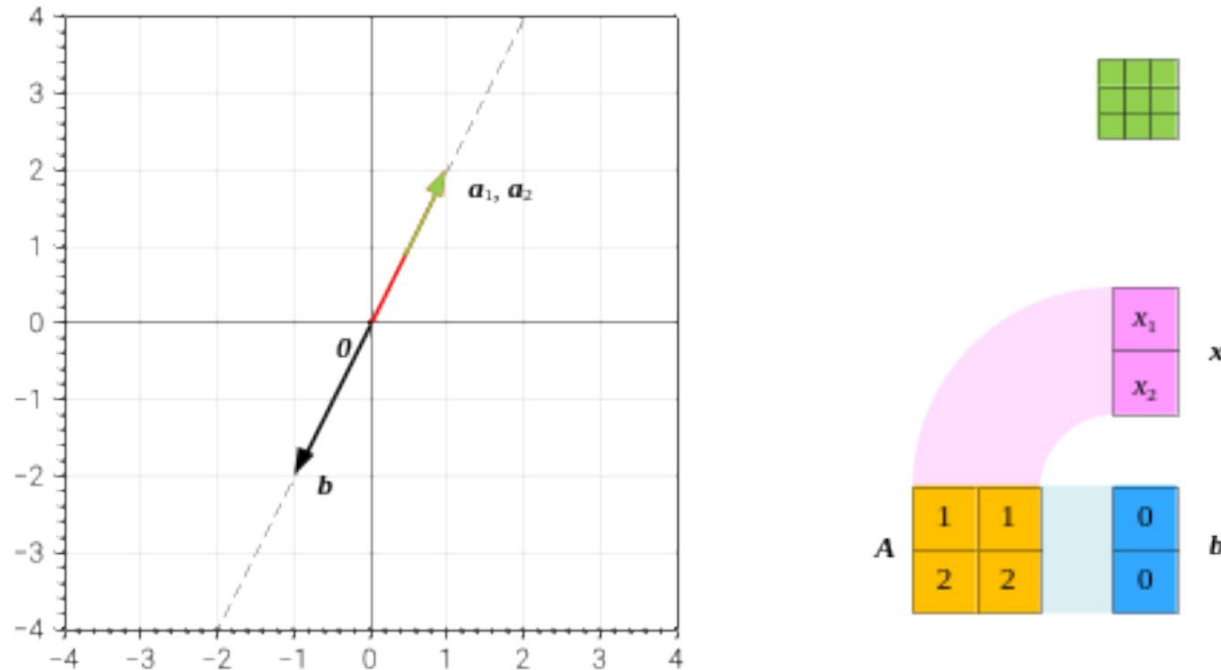


图 10. 方阵 A 行列式为 0, $Ax = b$ 无数组解

Same a_1, a_2 , but b Is Off the Line

$$\mathbf{b} = [-2, 0]^T$$

- Every linear combination of $\mathbf{a}_1, \mathbf{a}_2$ stays on the line through the origin, but $\mathbf{b} = [-2, 0]^T$ is not on it — no solution

Fig 11 · b Is Off the Line Spanned by a_1, a_2

- No matter how a_1, a_2 are combined, they can never reach a b that is off their shared line — no solution

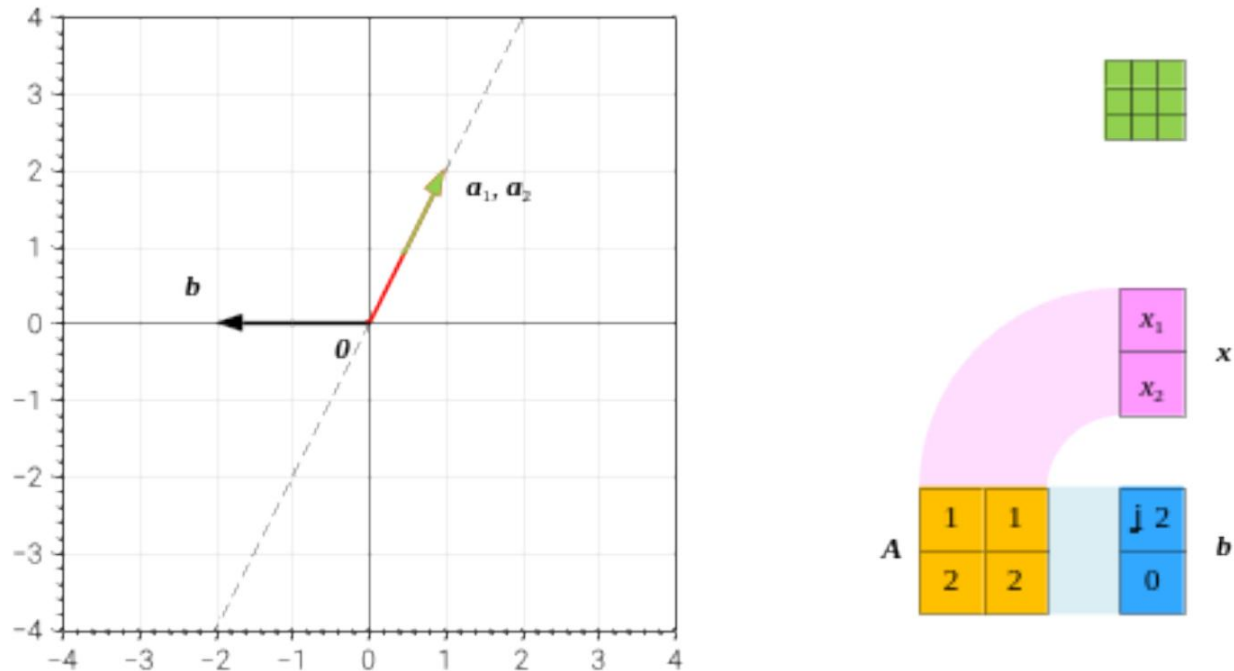


图 11. 方阵 A 行列式为 0, $Ax = b$ 无解

Same a_1, a_2, b Replaced by 0

$$k \times a_1 + (-k) \times a_2 = 0$$

- The zero vector 0 is obviously on the shared line; any real k satisfies the equation, so $(x_1, x_2) = (k, -k)$ are all solutions

$$AX=0$$

Fig 12 · Homogeneous System, Infinitely Many Solutions

- When a_1, a_2 are collinear, $Ax=0$ always has infinitely many solutions, with general form $(k, -k)$

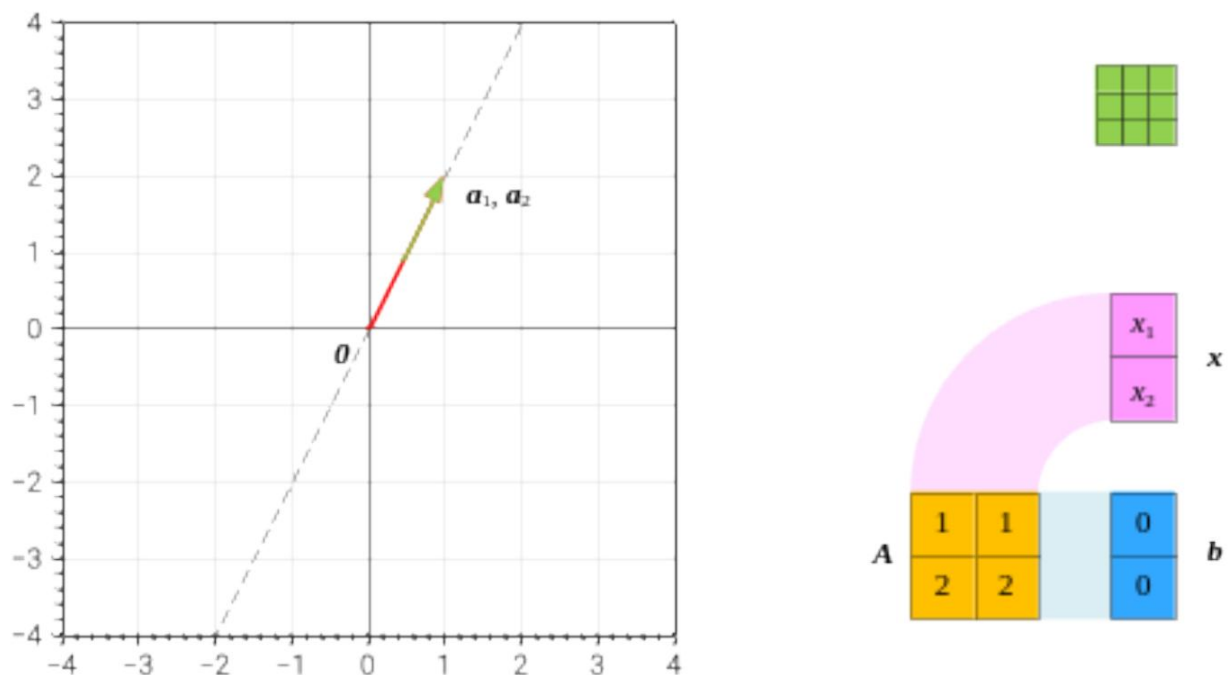


图 12. 方阵 A 行列式为 0, $Ax=0$ 无数组解

More Equations Than Unknowns

$$\mathbf{A}(m \times 2) \mathbf{x}(2 \times 1) = \mathbf{b}(m \times 1), m > 2$$

- With 3 linear equations in 2 unknowns, coefficient matrix $\mathbf{A}$ is 3×2 , $\mathbf{x}$ is 2-D, and $\mathbf{b}$ is 3-D
- Geometrically each equation is a line; the solution depends on how these lines relate to one another in the plane

Fig 13 · Four Overdetermined Systems, All Unsolvable

- (a) three parallel lines; (b) two coincide, one parallel; (c)(d) lines cross pairwise with no common point

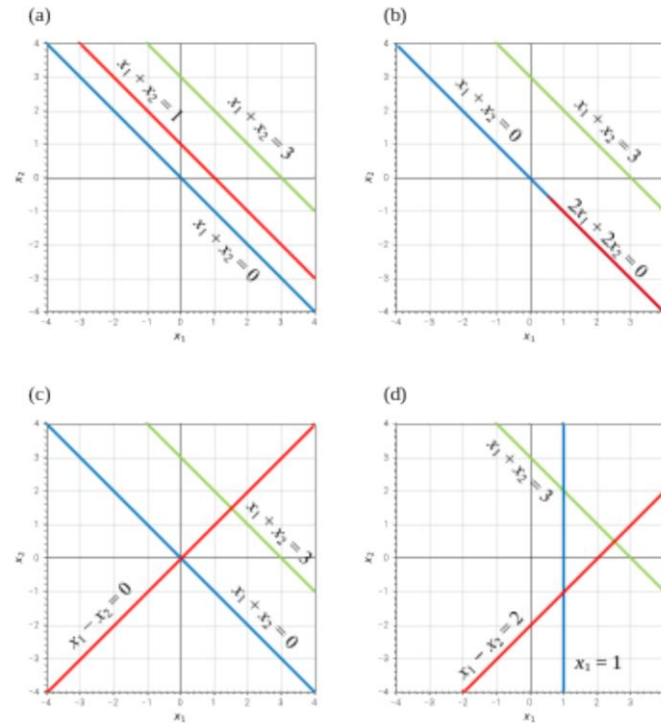


图 13. 超定方程组 (非齐次), 无解

$\mathbf{a}_1 = (1, 1, 1)$, $\mathbf{a}_2 = (1, 1, 1)$ Are Collinear

$$x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 = \mathbf{b} \Rightarrow \mathbf{b} = [1, 3, 0]^T$$

- $\mathbf{a}_1, \mathbf{a}_2$ are collinear, spanning only a line through the origin; $\mathbf{b}$ is off that line, so there is no solution

Fig 14 · a_1, a_2 Collinear, b Off the Line

- In 3-D, a_1, a_2 coincide into one line, while b strays off it — the system has no solution

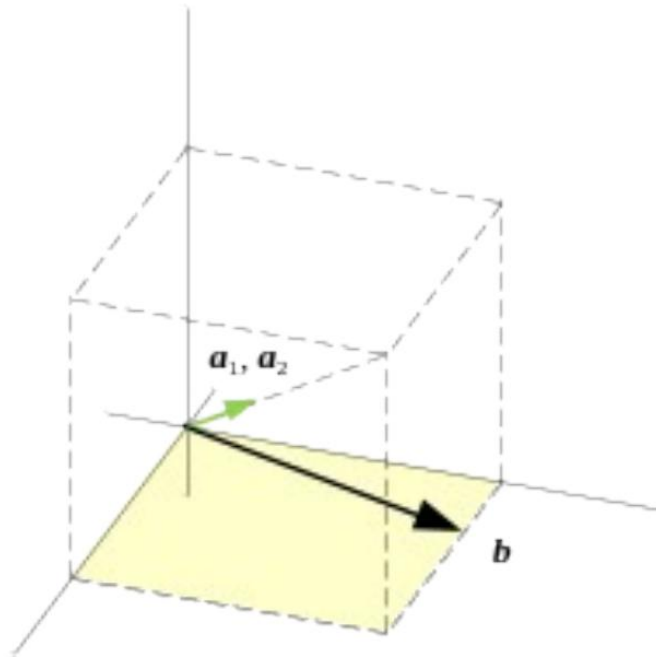


图 14. a_1, a_2 共线，而向量 b 不在这条 (过原点) 的直线上

Fig 15 · Three Lines Meet at One Point

- (a) two lines coincide and meet the third at a point; (b) three distinct lines meet at one point — both unique solutions

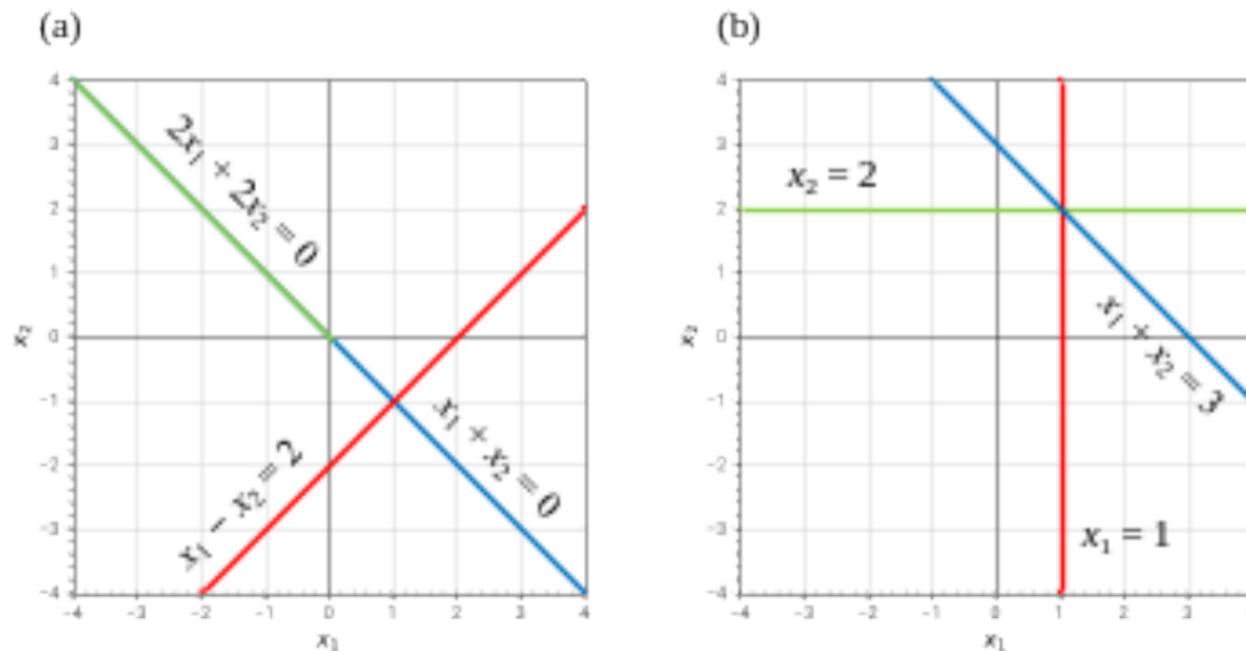


图 15. 超定方程组 (非齐次), 唯一解

$a_1=(1, 0, 1), a_2=(0, 1, 1)$ Are Not Collinear

$$1 \times a_1 + 2 \times a_2 = b \Rightarrow b = [1, 2, 3]^T$$

- a_1, a_2 are not collinear, but a_1, a_2, b are coplanar — b can be written as their linear combination

Fig 16 · a_1, a_2 Not Collinear, but Coplanar

- a_1, a_2 span a plane, and b happens to lie in it — hence the system has a unique solution

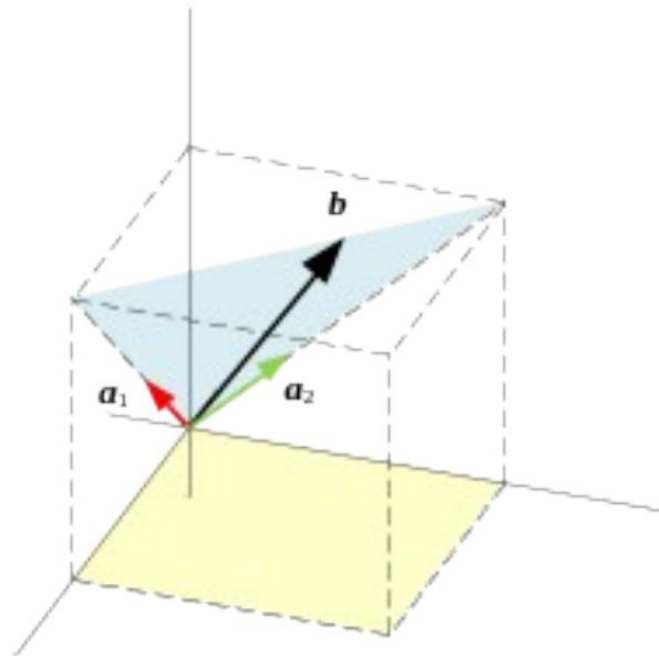


图 16. a_1, a_2 不共线, 但是 a_1, a_2, b 共面

Fig 17 · Two Homogeneous Systems, Solution at the Origin

- In both (a) and (b), all lines meet only at the origin — the unique solution is $\mathbf{x}=[0, 0]^T$

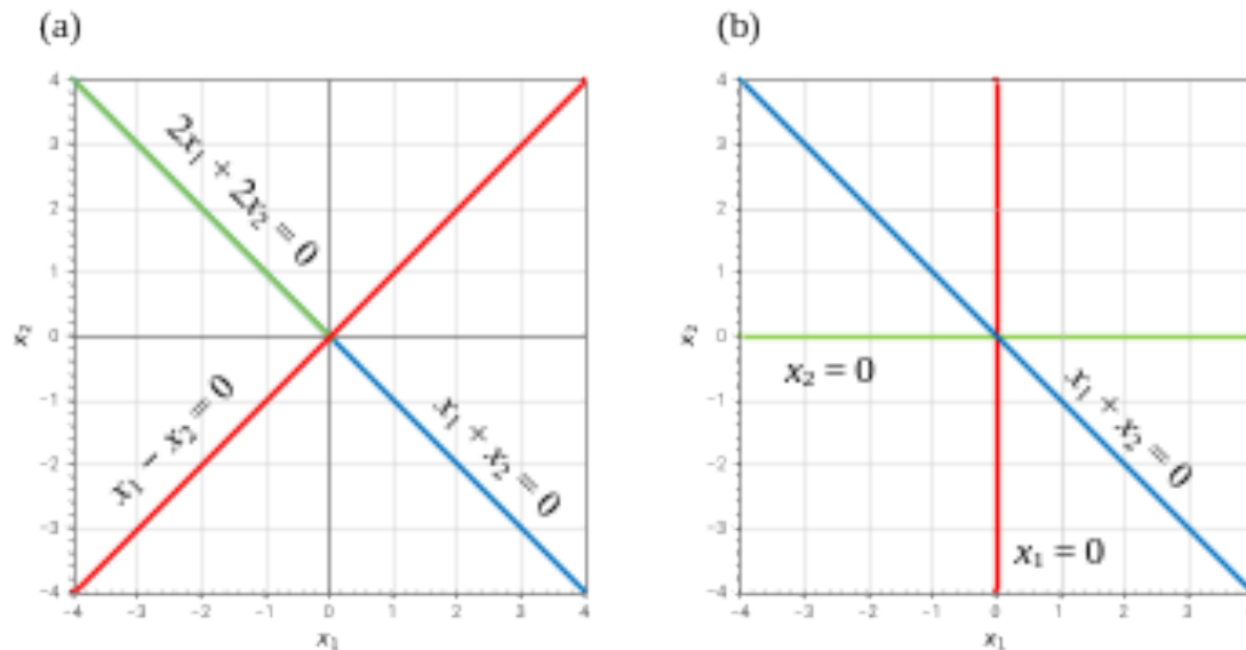


图 17. 超定方程组 (齐次), 唯一解

Fig 18 · Three Coinciding Lines, Infinitely Many Solutions

- (a) three lines coincide, representing the same line in different forms; (b) is its homogeneous counterpart

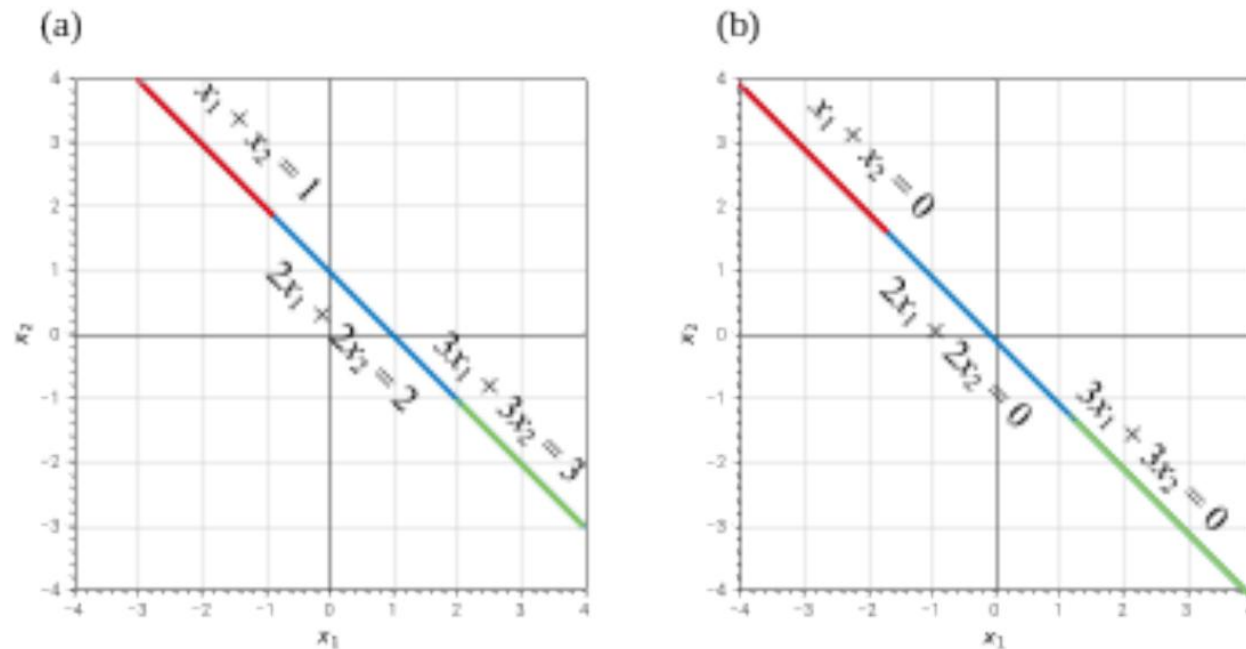


图 18. 超定方程组，无数组解

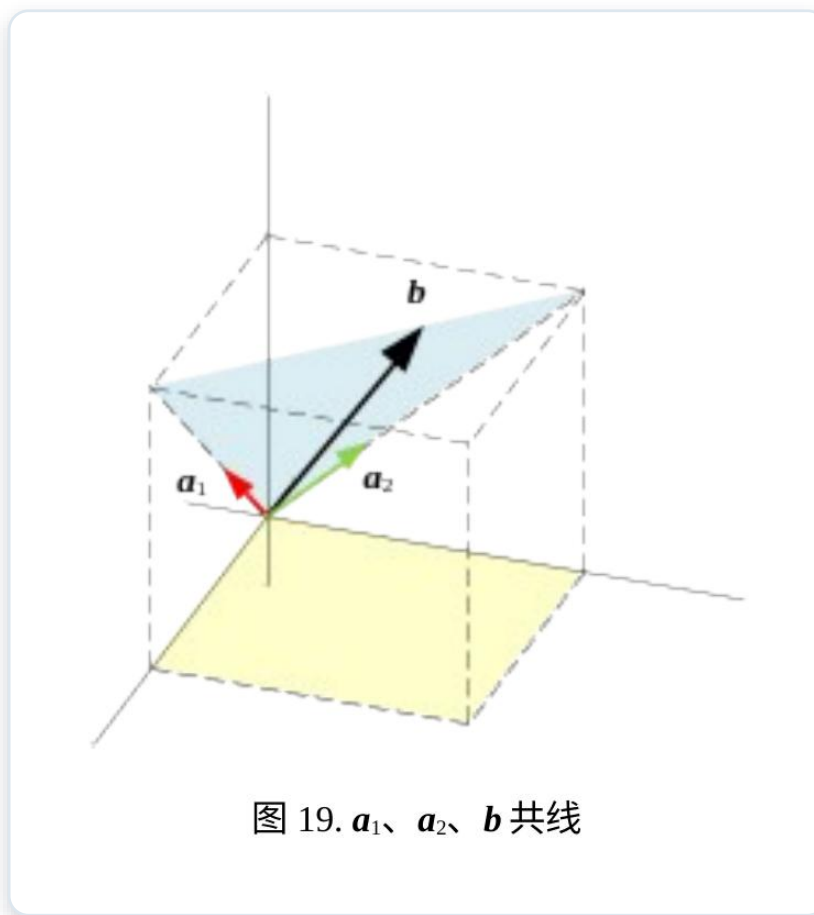
$a_1 = a_2 = (1, 2, 3)$, b Is Also on This Line

$$(k+1) \times a_1 - k \times a_2 = b$$

- a_1, a_2, b are collinear (in fact coincide); any real k satisfies the equation — infinitely many solutions

Fig 19 · a_1, a_2, b Are Collinear

- All three vectors coincide on one line; the combination coefficients $(k+1, -k)$ have infinitely many valid choices



***b* Replaced by *0*: Infinitely Many Solutions**

$$k \times \mathbf{a}_1 + (-k) \times \mathbf{a}_2 = \mathbf{0}$$

- $\mathbf{a}_1, \mathbf{a}_2$ are collinear; any real k makes the combination equal $\mathbf{0}$, so the homogeneous overdetermined system also has infinitely many solutions

From Line Intersections to Collinearity and Coplanarity

$\mathbf{a}_1, \mathbf{a}_2$ independent $\Leftrightarrow$ unique solution; dependent $\Leftrightarrow$ none or infinitely many

- The third perspective views $\mathbf{Ax}=\mathbf{b}$ as a linear combination of columns: whether $\mathbf{b}$ can be expressed via $\mathbf{a}_1, \mathbf{a}_2$ decides solvability
- Same logic for overdetermined systems: whether $\mathbf{a}_1, \mathbf{a}_2$ are collinear sets the basis's dimension; whether $\mathbf{b}$ lies within it decides solvability

Key Terms in This Section

Third Perspective $Ax=b$ viewed as a linear combination of A 's columns

Basis A set of linearly independent vectors spanning a space, e.g. $[a_1, a_2]$

Linearly Dependent Some vector in the set can be expressed via the others (e.g. collinear)

Homogeneous Linear System $Ax=0$

Collinear Multiple vectors lying on the same line through the origin

Linear Combination $x_1a_1+x_2a_2+\dots+x_pa_p$

Coordinates The coefficients (x_1, x_2) expressing b under basis $[a_1, a_2]$

Linearly Independent No vector in the set can be expressed via the others

Overdetermined System More equations than unknowns

Coplanar Multiple vectors lying in the same plane through the origin

FURTHER LEARNING

Extended Reading

Further Reading · `numpy.linalg.lstsq()`

LA_07_02_02.ipynb — least-squares approximate solutions for overdetermined systems

Further Reading · 3-D Vector Visualization

LA_07_02_03.ipynb — plotting vectors and grids in 3-D with matplotlib

Further Reading · Rank and Solvability

LA_07_02_04.ipynb — a systematic summary of solution criteria via `matrix_rank()`

Open-Source Repository

All code and figures for the book: <https://github.com/Visualize-ML>

Q1 – Q8

Q1 Write these two overdetermined systems as matrix products: $x_1+x_2=3$, $2x_1+4x_2=10$; $x_1=1$, $x_2=2$.

Q3 Use Python to verify that $\mathbf{b}$'s coordinates under basis $[\mathbf{a}_1, \mathbf{a}_2]$ are (1, -1) in Figs 3 through 12.

Q5 Plot the grids from Fig 3 (identity) and Fig 6 (upper triangular) side by side and compare their shapes.

Q7 Use `numpy.linalg.det()` to verify that matrix $A=[[1,1],[2,2]]$ in Figs 10-12 indeed has determinant 0.

Q2 Expand Q1's matrix products via the third perspective (linear combination of columns) and plot the grids.

Q4 Write a function `third_view(A, x)` that returns the terms of $A@x$ expanded as a column linear combination.

Q6 Use `numpy` to verify that $\mathbf{a}_1=[1,2]$, $\mathbf{a}_2=[-1,2]$ in Fig 8 are independent and span the whole plane.

Q8 Write code that decides whether two given 2-D vectors $\mathbf{a}_1, \mathbf{a}_2$ are collinear (linearly dependent).

Q9 – Q16

Q9 Using Q8's function, check whether $\mathbf{a}_1=\mathbf{a}_2=[1,2]$ in Figs 10-12 are collinear, and explain why that means the solution isn't unique.

Q11 Use numpy to verify Fig 11's $Ax=b$ ($b=[-2,0]$) truly has no solution — b is off the line spanned by a_1, a_2 .

Q13 Use `numpy.linalg.matrix_rank()` to compute the ranks of Fig 10's coefficient matrix A and augmented matrix $[A|b]$.

Q15 Build a 3×2 overdetermined coefficient matrix A , find its rank in Python, and state the rank condition for a unique solution.

Q10 Use numpy to verify Fig 10's $Ax=b$ ($b=[-1,-2]$) truly has infinitely many solutions — try 3 values of k .

Q12 Write a function `is_on_line(v, a)` that checks whether vector v is collinear with vector a (i.e. $v=k*a$).

Q14 Use `numpy.linalg.matrix_rank()` to verify that A in Fig 12 has rank 1, less than the 2 unknowns — hence infinitely many solutions.

Q16 Use matplotlib to plot the three parallel lines in Fig 13(a), confirming they share no common point.

Q17 – Q25

Q17 Solve the system behind Fig 15(b), computing the least-squares solution with `numpy.linalg.lstsq()` and comparing it to the exact solution.

Q19 Plot $\mathbf{a}_1$, $\mathbf{a}_2$, $\mathbf{b}$ from Fig 16 in 3-D with matplotlib, and verify whether they are coplanar.

Q21 Using Q20's function, check coplanarity for the three vectors in Fig 14 and in Fig 16, and compare the results.

Q23 Use numpy to verify Fig 18(b)'s $k \cdot \mathbf{a}_1 + (-k) \cdot \mathbf{a}_2 = \mathbf{0}$ holds for any k , and explain why this is the homogeneous general solution.

Q25 Pick any 3×2 overdetermined system, determine its solution type, and justify it with a third-perspective plot.

Q18 Use numpy to verify that both homogeneous overdetermined systems in Fig 17 have the unique solution $(0, 0)$.

Q20 Write a function `is_coplanar(a1, a2, b)` that uses a determinant to test whether three 3-D vectors are coplanar.

Q22 Use numpy to verify Fig 18(a)'s $(k+1)\mathbf{a}_1 - k \cdot \mathbf{a}_2 = \mathbf{b}$ holds for any k — test $k=0, 1, -1$.

Q24 Pick any invertible 2×2 matrix A and vector $\mathbf{b}$; plot both the matrix-multiplication view and the linear-combination view.

10 Prompts to Explore with AI

1 How does the “third perspective” of matrix multiplication differ from the row-by-row dot-product view?

2 Prove that if the vector set $[\mathbf{a}_1, \mathbf{a}_2]$ is linearly dependent, it cannot form a basis for the plane.

3 Explain the concepts of “basis” and “coordinates,” with a concrete 2-D example.

4 Use Python to demonstrate that the same solution (1, 2) can correspond to several different coefficient matrices $\mathbf{A}$.

5 Why does an overdetermined system (more equations than unknowns) usually lack an exact solution, but admit a least-squares approximation?

10 Prompts to Explore with AI

6

Explain how three lines being “concurrent,” “coincident,” or “parallel” corresponds to different solution types of an overdetermined system.

7

Explain “collinear” and “coplanar” geometrically, and how they relate to linear dependence.

8

Design a small Python project: given any 3×2 overdetermined system, automatically classify and visualize its solution.

9

Explain how the solution set of $\mathbf{Ax}=\mathbf{0}$ relates to whether $\mathbf{a}_1, \mathbf{a}_2$ are linearly independent.

10

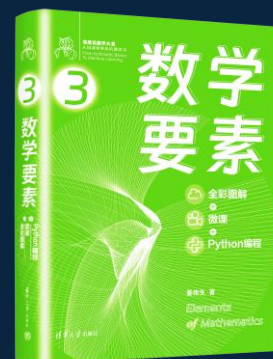
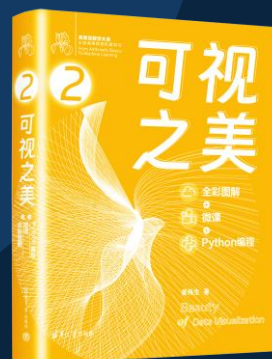
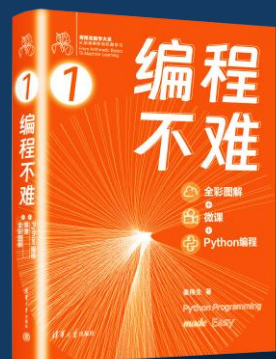
Compare this section's approach (third perspective / linear combination) with the previous section's (elimination / row operations).

数学

数学基础

线性代数

概率统计



Python编程

数据可视化

工具

数据分析
图和网络

回归、降维
分类、聚类

实践

